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RESEARCH ARTICLE

Are swimming fish dual oscillator systems? A case study using free-swimming smooth dogfish sharks (*Mustelus canis*)

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ABSTRACT

Fish swimming has classically been modeled as a rearwardly propagating wave of increasing amplitude and fixed frequency, based on kinematic data from large numbers of species in captivity. However, recent work on sharks swimming in natural environments has suggested that anterior and posterior body segments oscillate at different frequencies from each other. We attached accelerometer, gyroscope and magnetometer data loggers to the anterior and posterior body sections of smooth dogfish, *Mustelus canis* (*n*=4), and released these individuals in the wild. In over 25 h of recording, the anterior and posterior frequency estimates were within 0.1 Hz of each other for 97±1.9% of the time. Additionally, the phase differences between the anterior and posterior body regions were narrowly distributed, indicating that the anterior and posterior were phase locked at the same frequency, demonstrating that smooth dogfish are not dual oscillating locomotor systems. We highlight the fact that when comparing the frequency of oscillation along the body, emphasis needs to be placed on analyzing the phase difference between body parts with sufficient accuracy to test for significant differences, not just the dominant frequency.

KEY WORDS: Shark, Locomotion, Kinematics, Oscillation

INTRODUCTION

In order to swim, most fish rhythmically bend their bodies, which induces a lateral movement tangential to their swimming direction (Webb, 1975; Donley and Dickson, 2000; Long et al., 1994; Jayne and Lauder, 1995). By altering the timing of when each body segment bends, the lateral movement appears as a rearwardly propagating wave of increasing amplitude (Smits, 2019; Jayne and Lauder, 1995; Gray, 1933; Shadwick and Goldbogen, 2012). The motion of the body can be described as a function of three terms: the frequency of the tailbeat, the phase lag (determining the wavelength), and the lateral amplitude at each part of the body (Videler and Hess, 1984). These parameters provide a quantitative framework to describe body kinematics, allowing for comparisons and groupings across species. (Breder, 1926; Di Santo et al., 2021). While there are minor variations in wavelength or amplitude within a species during steady locomotion, tail beat frequency can vary by an order of magnitude, and is the primary mechanism of speed modulation for fish species (Bainbridge, 1958; Floryan et al., 2017). Regardless of the temporal variations in tailbeat frequency, the kinematics of the body are based on the same frequency.

In most experimental descriptions of fish swimming kinematics and in mathematical analyses of the kinematic waveform (Akanyeti et al., 2022), it is assumed that the whole body experiences the same fundamental frequency. However, additional frequencies are needed to accurately and comprehensively describe the kinematics of different body segments (Root et al., 2007; Long et al., 2010). These contributions can be from minor frequencies which are harmonics of the tailbeat frequency, or they can be ‘aperiodic’ or ‘secular’, which are of increased importance during unsteady swimming (Root et al., 2007).

The presence of multiple frequencies along the body has recently been proposed to occur in several shark species (*Sphyrna lewini*, *Sphyrna tibero*, *Carcharhinus perezi*, *Carcharhinus limbatus*), when swimming freely in the wild or under near natural conditions (Hoffmann et al., 2017; Porter et al., 2020; Kajiura et al., 2022). In these four species, it was found that the yawing frequency at the anterior part of their body was significantly faster than the tailbeat frequency, and these fish were deemed to be dual oscillating systems. These data suggest that even though the body functions as a single flexible bending beam, different regions such as the head and tail could be operating at different frequencies. This concept that the anterior and posterior regions of the fish body are operating at different frequencies directly contradicts the common methods used to describe fish kinematics, which assume a single oscillatory frequency. These multi-frequency kinematics have been primarily identified in fishes with backbones composed of calcified cartilage and it has been suggested that different oscillation frequencies between the head and tail could be a common kinematic phenomenon in elasmobranch species (Porter et al., 2020). As these kinematic data were obtained from sharks swimming freely and not confined to laboratory flow tanks or space-limited aquaria, the phenomenon of dual oscillation frequencies in a single bending body could have been overlooked in previous laboratory analyses of fish swimming kinematics.

However, in the shark species analyzed to date, the frequency of the head was not a harmonic of the tail beat frequency. The reported head to tail frequency ratios for each of the four shark species were: 1.7:1 Hz, 1.6:1.2 Hz, 0.62:0.53 Hz, 1.23:0.82 Hz (Hoffmann et al., 2017; Porter et al., 2020; Kajiura et al., 2022). To fully describe the kinematic envelope and estimate variation in head versus tail oscillation frequency, multiple tailbeats have to be observed which requires an individual to be within the camera frame for an extended time. If only a relatively short period of 3–6 tailbeats can be observed, the possibility of behavioral and kinematic variation and measurement error can render it challenging to accurately quantify oscillator frequency, especially when using traditional video-based methods during unsteady behaviors and relatively low sampling rates (Root et al., 2007). If anterior and posterior body locations show frequency differences during steady swimming, we would

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also expect to see varying phase shifts between the anterior and posterior of the body.

We illustrate the type of data and analyses needed to distinguish between dual and single oscillator systems using a simulation shown in Fig. 1. For a dual oscillator system where the anterior and posterior regions of the body operate at different frequencies, the expectation is that the difference in phase of each body part will vary through time and a circular plot will show a range of phases (Fig. 1C). This contrasts with the fish body operating as a single oscillator system where both the anterior and posterior body regions operate with the same

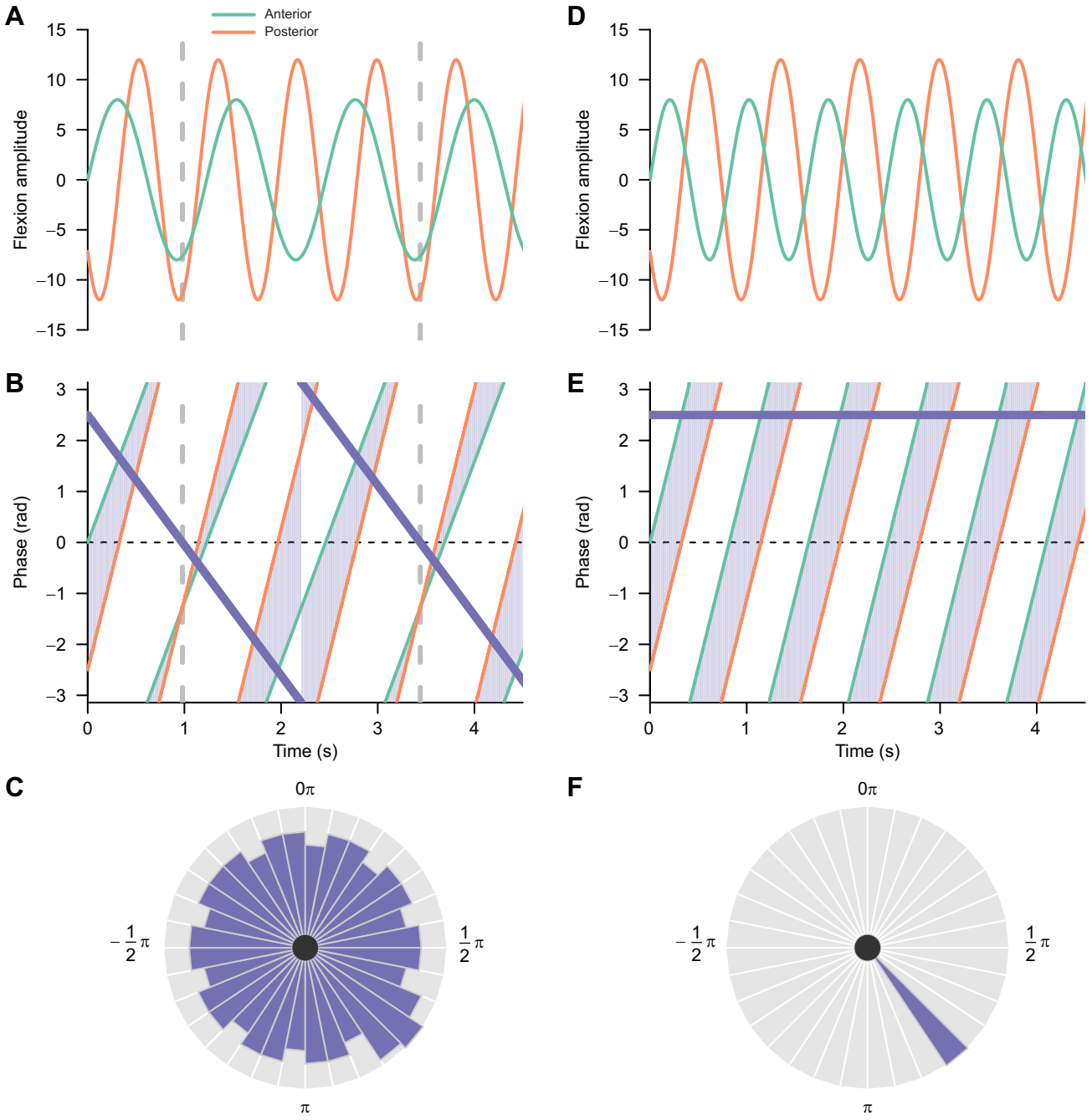

Fig. 1. Simulations comparing dual oscillating and single oscillating systems. Time-varying patterns in yawing amplitude (A,D) and the phase of movement (B,E) for the anterior (green) and posterior (orange) body segments. In A and B, frequency estimates were taken from Porter et al. (2020), with the anterior of the body oscillating at 1.23 Hz and the posterior of the body at 0.83 Hz to show the time varying phase for each body part (B) and phase difference (purple) between the body segments expected under the hypothesis of dual oscillating body kinematics. With two stable frequencies, the phase difference is time varying, resulting in a uniform distribution as shown in a circular histogram (C). (D,E) The traditional model, with the anterior and posterior segments displaying the same frequency (0.82 Hz) and as a result the phase difference is static, representing the consistency in frequency between the anterior and posterior regions of the body. The distribution in phase over time is shown as a circular histogram with a uniform distribution for the dual oscillating system (C), and a highly clumped distribution for the single oscillating system (F). A phase difference of 0π would indicate that both body segments are in phase, while a phase difference of π would indicate that the two body parts are in antiphase. Vertical grey dotted lines represent the duration of a fundamental frequency that contains the pattern between the two body parts. Each circle in C and F is divided into π/16 sections.

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frequency and they maintain a near constant phase difference. Challenges in accurately measuring body kinematics and measurement error may introduce a small amount of variation in such phase plots, but analyses of this type will permit a clear discrimination between single and dual oscillator systems in fish locomotor kinematics.

To study kinematic harmonics in greater depth for longer periods of time with a specific focus on sharks as possible dual oscillating systems, we attached acceleration, angular velocity and magnetic field data loggers to sharks at three locations on their body (anterior, mid, posterior) and released them into the wild. These loggers allow us to continuously record the frequency of swimming kinematics over long periods of time and from multiple locations simultaneously (White and Lauder, 2024). Our aim is to test the dual-oscillator hypothesis: do we observe different frequencies between the anterior and posterior body segments over longer periods of time? A second goal was to determine if variation in observed frequencies generates a shifting pattern in the phase differences as would be expected from a dual-oscillator system (Fig. 1C).

MATERIALS AND METHODS

In September and October 2023, four healthy smooth dogfish [*Mustelus canis* (Mitchill 1815); Table 1] were acquired from the Marine Biological Laboratory (Woods Hole Massachusetts, USA). Sharks were anesthetized in Syncaine (MS-222, Syndel, Ferndale Washington) and a multi-datalogger package was attached to each shark (Fig. 2) at three locations: the anterior body (dorsal surface, anterior to the gill slits), mid body (posterior to the first dorsal fin) and posterior (on the lateral surface at the caudal peduncle). Each data logger was affixed to the shark using a suture at both the anterior and posterior of each tag and held in place with a galvanic timed release (International Fishing Devices Inc., Northland, New Zealand). Total time anesthetized was under 20 min for all individuals, and individuals were allowed to recover after surgery for 20 min. After tag attachment, all sharks regained equilibrium and swimming behavior and then were released into Cape Cod Bay, either from near shore or slightly offshore (<1 km) and allowed to swim at large (Fig. 2). Tag attachment and data acquisition was conducted under approved animal protocol IACUC 20-03-4 by Harvard University and in accordance with local regulations.

The multi-datalogger package included three dataloggers (AGM model, TechnoSmart Inc, Rome, Italy), with a 100 mAh battery (overall dimensions, 14×24×9 mm; mass, 6 g). Each data logger recorded at 100 Hz from a 3D accelerometer, 3D gyroscope and 3D magnetometer, and recorded temperature and pressure at 1 Hz. As a consequence of the high sampling rate, and inclusion of a gyroscope and magnetometer, the estimated battery life for each of these tags was 10 h. The three data loggers were attached to each other using 100 lb (45.3 kg) mono filament line. The anterior and posterior tags were encased in a small 3D printed body to hold the galvanic time release and to allow for a secure attachment point for the monofilament line. The mid-body tag was incorporated into a custom float package which also contained a VHF pinger (Advanced Telemetry Systems, Isanti MN; model: MM130). Each of the three tags were tethered in place by a galvanic timed release, designed to corrode over 12 h, releasing the dataloggers from the shark. After the dataloggers detach, the mid-body float brought the group of three loggers to the surface (Whitmore et al., 2016). This datalogger package was then located and recovered at sea using the VHF transmitter.

Once recovered, data from the three loggers was individually downloaded and analyzed. Each dataset was trimmed to the period before the tags detached from the shark. The point of detachment of each individual logger was identified as when the tag experienced a

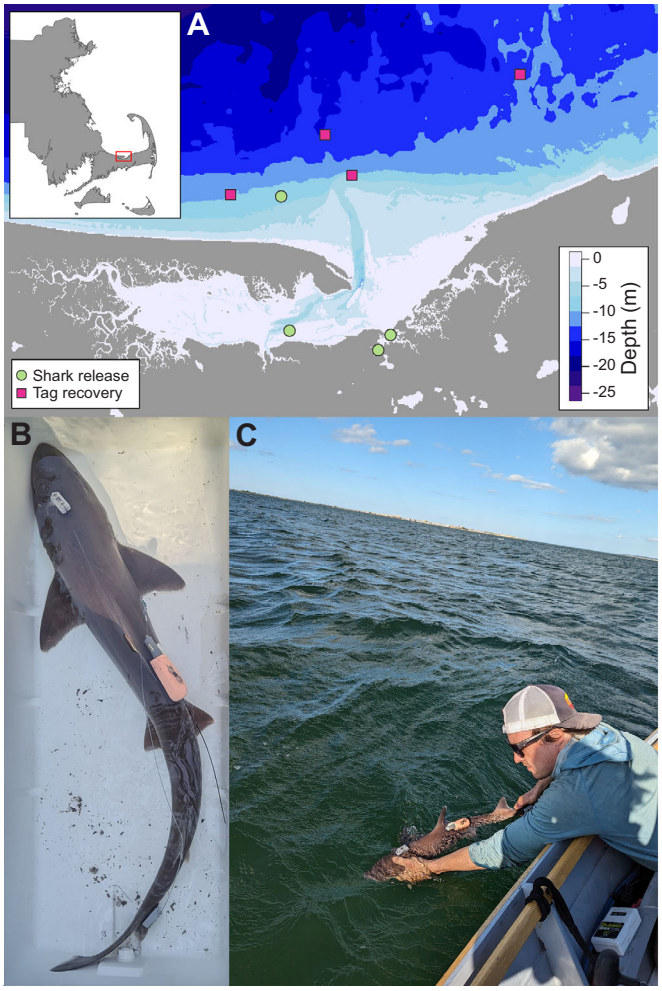

Fig. 2. Tagging and release of dogfish shark (*Mustelus canis*). (A) Bathymetry map of Barnstable Harbor, Massachusetts, USA, displaying the shark release locations (green circles) and the tag recovery locations (red squares). The inset map shows the study location (red box) in reference to the state of Massachusetts. Owing to large tidal movements and the number of hours between tag detachment and tag recovery, recovery locations are not detachment locations. (B) Shark 2 immediately before release: the anterior and posterior tags are the small grey squares attached to the shark and the orange object behind the first dorsal fin is the float containing the VHF transmitter. The middle body tag is visible as the small grey square next to the orange float. (C) After tagging, sharks were released into Cape Cod Bay to display volitional locomotor behavior over a 10 h period.

Table 1. Shark deployment information and frequency

Individual	Total length (m)	Recording duration (h)	Tag position	Peak *f* (Hz)	*f* (Hz; mean ±s.d.)
Shark 1	0.85	8.98	Anterior	0.702	0.78±0.18
			Posterior	0.702	0.78±0.18
Shark 2	0.83	5.91	Anterior	1.14	1.08±0.2
			Posterior	1.14	1.09±0.2
Shark 3	1.15	3.69	Anterior	0.71	0.78±0.23
			Posterior	0.71	0.78±0.23
Shark 4	1.00	7.99	Anterior	0.98	0.96±0.14
			Posterior	0.98	0.96±0.14

Frequency estimates are based on results from the continuous wavelet analysis.

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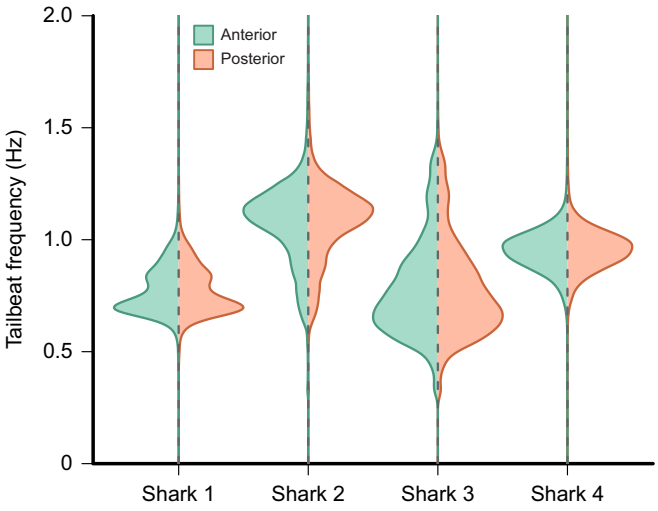

Fig. 3. Distribution of tailbeat frequencies for each shark. Frequency estimates are based on the dominant wavelet frequency when the shark was actively swimming and is based on each individual's entire deployment. The anterior (green, left) and posterior (orange, right) body frequencies are displayed for each shark and mirror each other showing similar distributions in tailbeat frequency.

substantial change in pitch or roll over a sustained period of time and was associated with larger amplitude acceleration and angular velocity values (Whitney et al., 2018; Zhang et al., 2022; Johnson and Tyack, 2003). These patterns were indicative of the tag detaching from the shark and dangling by the monofilament line while another logger was still attached and recording routine values. Magnetometer data were corrected for hard iron offsets by fitting a sphere to the raw data (Bidder et al., 2015). All tag measurements were corrected for their attachment angles and rotated so that the tag *x*-, *y*- and *z*-axes, aligned with the longitudinal, lateral and vertical axes of the shark's body, respectively (Jorgensen et al., 2015; Whitney et al., 2018). Each sensor (accelerometer, gyroscope and magnetometer) provides information on the movement and attitude of the tag, but for accurate 3D orientation (pitch, roll, yaw) estimates these sensors need to be fused together (Madgwick, 2010). In this study, a complementary filter was used to fuse the magnetometer, gyroscope and accelerometer data together to generate point estimates of pitch, roll and yaw (Mahony et al., 2005; White, 2025).

We estimated the frequency in the anterior and posterior tags using a continuous wavelet transformation to decompose the rate of yaw change (gyroscope-Z) into a frequency (87 levels between 4 and 0.3 Hz) and amplitude spectrum with the *biwavelet* package (Gouhier et al., 2013) in R (r-project.org). Wavelet analysis allows for spectrum estimates with respect to time and frequency and thus excels at analyzing signals with temporally varying frequencies (Marchant, 2003; Torrence and Compo, 1998). At each time point in the dataset, the dominant frequency was selected as the frequency with the greatest signal power. In addition to the wavelet analysis, we used a peak tracking algorithm in the *pracma* package (https://CRAN.R-project.org/package=pracma) to find the local maxima

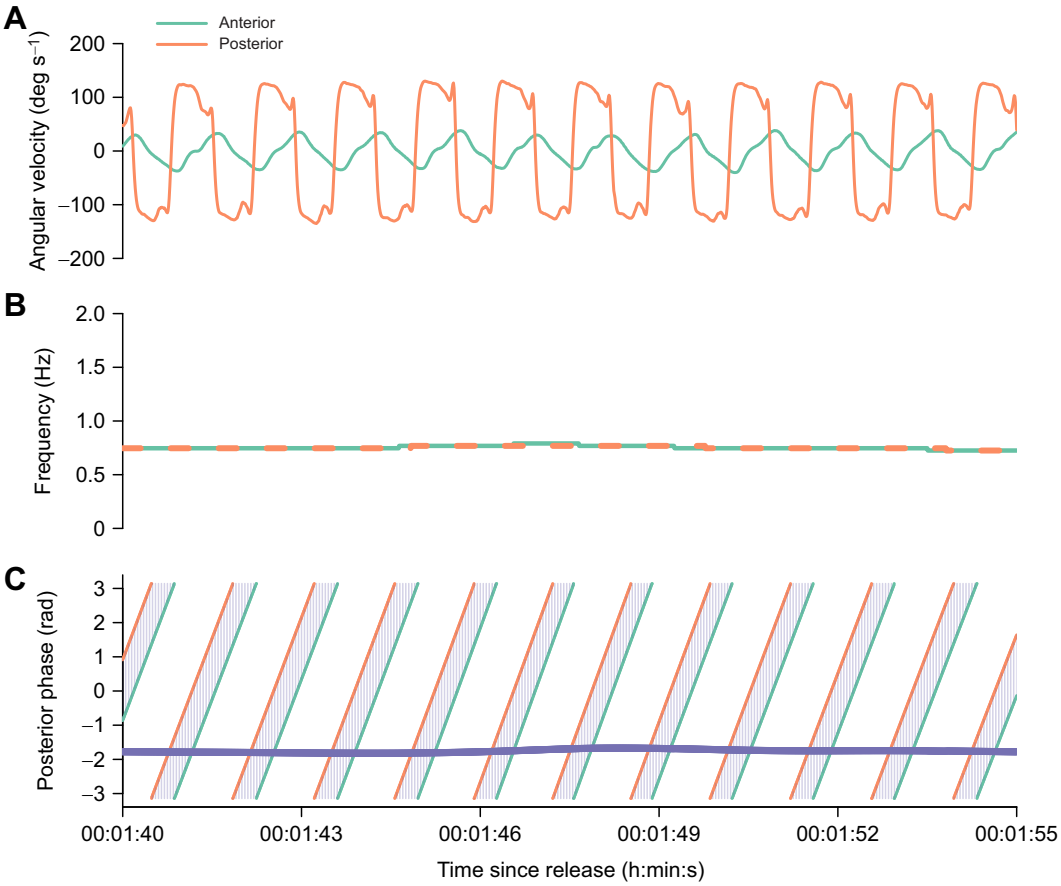

Fig. 4. Comparison of the anterior and posterior angular velocity, frequency and phase during a period of steady swimming. (A) Angular velocity, (B) frequency and (C) phase during a period (15 s) of steady swimming for shark 4. The phase difference between the anterior and posterior body segments is shown as the purple line and is nearly constant.

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and minima in the yaw data. The time from one peak to the next peak that contained a minima was used to isolate the duration of an individual tail beat. As tags record at a fixed frequency, our temporal resolution for determining periodicity is 0.01 s. However, this means that the resolution of our frequency estimates are variable and dependent on the measured frequency. For instance, a 0.5 Hz signal is characterized by 200 data points, providing a frequency resolution (±0.01 s) of 0.002 Hz. However, a 2 Hz tailbeat is characterized with 50 data points, and thus would have a frequency resolution (±0.01 s) of 0.04 Hz. This method provides an alternative method of measuring frequency which is comparable to other studies and allows for the statistical unit to be a tailbeat.

To identify whether the differences in anterior and posterior frequency are stable relationships, we examined the extent of variation in the phase shift between the anterior and posterior sections of the body. In addition to estimating the wavelet power, a continuous wavelet transform estimates the wavelet phase ($-\pi$: π) continuously through time. We extracted the phase of the dominant wavelet of both the anterior and posterior body and calculated the anterior–posterior phase difference. If individuals exhibited stable alternative frequencies between the head and tail regions, we would expect to see a relatively uniform distribution of phase differences (e.g. Fig. 1C). To not conflate stable locomotion with unstable dynamic behaviors by the sharks, steady state forward locomotion was isolated as times (>10 s) where the mean heading changed by less than 10 deg over 10 s. Rayleigh tests of uniformity were used to determine if the distribution of phase lags were significantly different than uniform, as would be expected if the anterior and posterior displayed stable and different frequencies.

RESULTS

Owing to minor variation in the galvanic timed release, attachment location and associated stressors, each tag detached from the shark at slightly different times. In all sharks, at least one tag detached before the tag's battery died. In two deployments, all three dataloggers detached from the shark and floated to the surface (shark 3, 9.5 h; shark 4, 9.4 h). In the other two deployments at least one datalogger was still attached to the sharks when all the tags batteries were depleted (shark 1, 11.1 h; shark 2, 10.6 h). We were able to record a total of 25.7 h of swimming behavior when each data set was limited to the time period before the first tag detached. All individuals spent the majority of the time swimming, and based on their depth data, were largely benthically associated and used habitats as deep as 17 m. However, shark 1 spent a considerable amount of time (45%) in the water column, based on its variation in depth.

Based on frequencies derived from the wavelet analysis, the smallest individual displayed the highest mean tailbeat frequency of 1.08 Hz, and the largest individual displayed the lowest tailbeat frequency (0.78 Hz). The anterior and posterior tags displayed the same peak and average frequencies (Table 1) and there were no clear differences in the distribution of anterior and posterior frequencies for any of the sharks (Fig. 3). When compared using

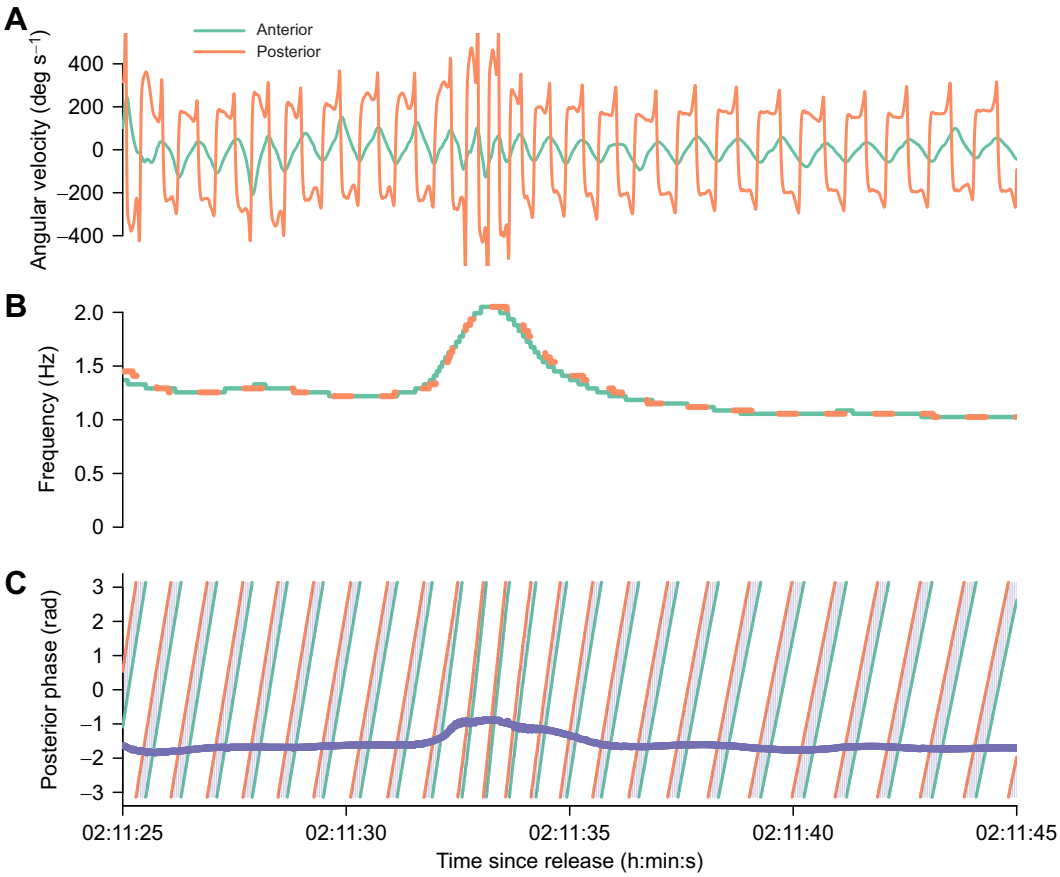

Fig. 5. Steady phase shift during a time of variable tailbeat frequency. Comparison of the anterior (green) and posterior (orange) angular velocity (A), frequency (B), and phase (C) during a period when shark 4 temporarily accelerates and increases its tailbeat frequency. The phase difference between the anterior and posterior body segments is shown as a purple line. There is a slight shift in phase during the acceleration, but this shift is only 5% of the anterior frequency.

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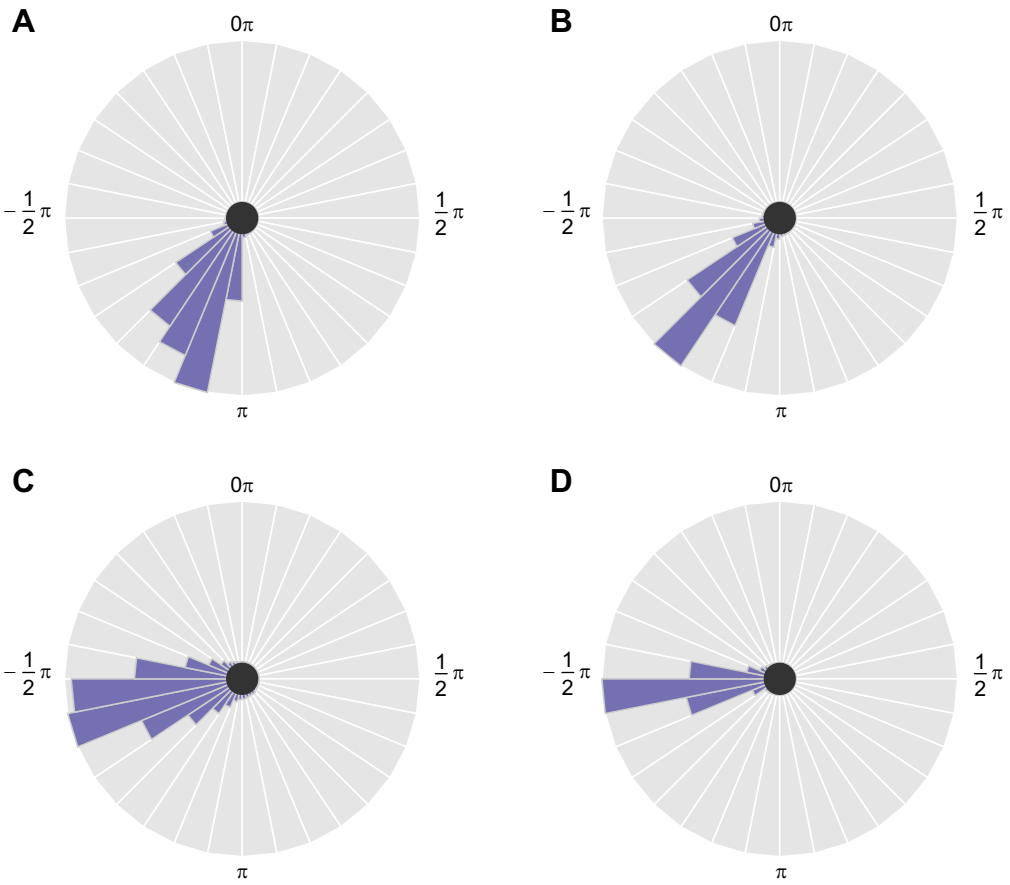

Fig. 6. Nonuniform distributions in anterior-posterior phase difference. Circular histograms of the phase difference (purple bars) between the anterior and poster body segments for the four sharks (A–D, respectively). The full circle is divided into π/16 sections, and each distribution has been scaled to the most frequently observed phase difference. A phase difference of 0π would indicate that both body segments are in phase, while a phase difference of π would indicate that the two body parts are in antiphase. The highly clumped phase pattern for each shark supports the single oscillator model of undulatory kinematics.

point-by-point estimates of frequency, the anterior and posterior frequency estimates were highly correlated (shark 1: r=0.97; shark 2: r=0.82; shark 3: r=0.88; shark 4: r=0.81), but the anterior and posterior tags produced the exact same frequency estimates only 75%, 60%, 43% and 68% of the time. The average difference between the two frequency estimates, when there were differences, was very small (0.007, 0.024, 0.030, 0.014 Hz; sharks 1–4, respectively) and 97.5±1.9% of the time produced frequency estimates that were within 0.1 Hz of each other.

Over a complete tail beat cycle, the distribution in phase (−π: π) should be uniformly distributed for an individual tag. All anterior and posterior tags were found to have uniform distributions (Rayleigh test statistic <0.001, P>0.1). If the two body parts display the same frequency, the phase lag between them should be constant (Fig. 1F). But if they exhibit different and stable frequencies, then the phase lag should be time varying with an approximately uniform distribution (Fig. 1C). When plotted over time, the phase lag between the two tags was approximately the same (Fig. 4) and this is true even as individuals change their tail beat frequency (Fig. 5). When compared over the entire data set, the phase difference between tags was found to have a nonuniform distribution (Fig. 6; shark 1: Rayleigh=0.87, P<0.001; shark 2: Rayleigh=0.52, P<0.001; shark 3: Rayleigh=0.46, P<0.001; shark 4: Rayleigh=0.949, P<0.001) and was highly directional, which is consistent with phase locking between the anterior and posterior tags.

To directly compare individual tail beats, we used a peak tracking algorithm. This method enabled us to directly measure each shark's frequency based on a large number of tailbeats (shark 1: 24,516; shark 2: 19,329; shark 3: 8374; shark 4: 23,194). The difference between the average anterior and posterior body regions yawing frequency was 0.05, 0.01, 0.01, 0.004 Hz, respectively, with the anterior body position displaying a lower mean frequency than the posterior. However, this represents a small percentage (1.6±3.2%) of the shark's average cycle time and when only tailbeats during steady swimming are analyzed, there was no difference in frequency between the two body positions, and the average periods were less than 0.01 s different.

DISCUSSION

In over 24 h of continuously monitored *in situ* volitional swimming kinematics in four individuals of the smooth dogfish shark (*Mustelus canis*), we did not observe any substantive differences in the frequency between the anterior and posterior sections of the body, suggesting that this species is not a dual-oscillating system. While instantaneous frequencies between the anterior and posterior body regions could deviate from each other, these deviations were very small and, based on the phase lag, indicate minor errors in estimation and not stable deviations in frequency. Previous work in the smooth dogfish has also documented that individuals in captivity display the same head and tail oscillation with a near 1:1 frequency (O'Connell et al., 2020). We did observe minor shifts between the anterior and posterior tags during the deployments, but these were on the same order of magnitude and direction as that expected from temporal drift between loggers (White and Lauder, 2024). Furthermore, the rate at which this drift occurred (0.2 s h^{-1}) would signal a difference in frequencies of less than 0.01 Hz.

Of the three separate studies that have reported differences in anterior and posterior frequency in sharks, only Porter et al. (2020) report a phase shift. This reported value is a static point estimate and not variable as would be expected if the anterior and posterior are

operating on different frequencies; however, these authors do note the large standard deviation around this estimate, which would be expected in an asynchronous system (Fig. 1). Across most swimming animals, the degree of lateral oscillation and postural reconfiguration is significantly reduced at the anterior body compared with the posterior region of the body (Long et al., 2010; Lauder and Tytell, 2005; Di Santo et al., 2021). As the head yawing amplitude during shark locomotion is of a reduced magnitude compared with that of the tail, accurately measuring small values of side-to-side oscillation using video can be difficult. This is especially true under variable field conditions with changing light levels, surface reflections and wave motion, all of which can introduce artifacts into kinematic estimates, ultimately decreasing the signal to noise ratio. Furthermore, to cover an area large enough to sufficiently capture volitional behavior, the effective resolution of the animal is decreased, ultimately reducing tracking resolution. Within a tailbeat cycle, minor harmonics have been used to increase the accuracy of modeling the lateral and axial movements of body segments (Root et al., 2007). When multiple frequencies are present, care should be taken to measure multiple tailbeats over a substantial amount of time and to isolate the multiple frequencies present. The patterns in the angular velocity of both the anterior and posterior of the body presented here deviate from the classical sine wave pattern, often used to model oscillatory locomotion.

Amongst animals there is wide variation in swimming kinematics, with strong links to morphology and life history strategy (Vogel, 2020; Smits, 2019). The previous studies that have documented dual oscillation have examined species of sharks that differ markedly from the smooth dogfish studied here. Smooth dogfish are non-obligate ram ventilators, can rest on the seafloor and are predators of benthic invertebrates (Gelsleichter et al., 1999; O'Connell et al., 2020; Bigelow et al., 1948). In contrast, hammerheads, blacktip sharks, and Caribbean reef sharks are generally considered to be obligate ram ventilators with a more pelagic distribution and are higher trophic level predators (Wells et al., 2018; Bowers and Kajiura, 2023; Shipley et al., 2018; Bigelow et al., 1948). Based on differences in diet and habitat, these species could have different selective pressures on their swimming kinematics with a larger focus on maximum performance (swimming speed), differing sensory perception or locomotor efficiency (Langerhans and Reznick, 2010). Accompanying these life history traits, are external and internal morphological differences that are known to vary by ecomorphotype which can impact an animal's kinematics (Porter et al., 2009; Hoffmann et al., 2020; Ingle et al., 2018). Finally, the sharks employed in this study had been handled immediately prior to their release, which can alter swimming behavior (Whitney et al., 2016). However, here we show that four smooth dogfish never employed a sustained period where the anterior and posterior body segments displayed different yawing frequencies in over 27 h of free swimming in the field, and our data strongly support the single oscillator model of undulatory kinematics. Future work is needed to determine the occurrence and prevalence of frequency differences along the body among diverse shark species, and the conditions under which it might be present.

By attaching dataloggers to multiple locations on the sharks, we were able to continuously monitor the movements of individuals over periods of hours, providing temporally and spatially unbiased estimates on animals *in situ* and generating large data sets of undulatory kinematics of over 40,000 individual tail beats. Using this method, we show that the entire body of the smooth dogfish has one fundamental primary frequency during locomotion – the tailbeat frequency – and there is no varying phase shift between body sections. Furthermore, we were able to measure this behavior when sharks were under natural swimming conditions for long periods of time, circumventing the limitations of traditional camera tracking and the constraints involved in traditional laboratory analyses of fish locomotor biomechanics.

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Competing interests
The authors declare no competing or financial interests.

Author contributions
Conceptualization: C.F.W., G.V.L.; Formal analysis: C.F.W.; Funding acquisition: G.V.L.; Methodology: C.F.W.; Project administration: G.V.L.; Supervision: G.V.L.; Visualization: C.F.W.; Writing – original draft: C.F.W., G.V.L.; Writing – review & editing: C.F.W., G.V.L.

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Data availability
All tag data sets used in this study are available upon request from the authors.

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